# ASCE 41-23 Tier 1 Checklists

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| FIRM: | |
| PROJECT NAME: | |
| SEISMICITY LEVEL: | |
| PROJECT NUMBER: | |
| COMPLETED BY: | |
| DATE COMPLETED: | |
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# 17.14CP Structural Checklist for Building Types PC1: Precast or Tilt-Up Concrete Shear Walls with Flexible Diaphragms and PC1a: Precast or Tilt-Up Concrete Shear Walls with Stiff Diaphragms

**Table 17-28. Collapse Prevention Structural Checklist for Building Types PC1 and PC1a.**

| Status | | | | Evaluation Statement | Tier 2 Reference | Commentary Reference | Comments |
|---|---|---|---|---|---|---|---|
| **Low Seismicity** | | | | | | | |
| ***Connections*** | | | | | | | |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | WALL ANCHORAGE: Exterior concrete or masonry walls that are dependent on the diaphragm for lateral support are anchored for out-of-plane forces at each diaphragm level with steel anchors, reinforcing dowels, or straps that are developed into the diaphragm. Connections have strength to resist the connection force calculated in the Quick Check procedure of Section 4.4.3.7. | 5.7.1.1 | A.5.1.1 | |

| Status | | | | Evaluation Statement | Tier 2 Reference | Commentary Reference | Comments |
|---|---|---|---|---|---|---|---|
| **Moderate Seismicity (Complete the Following Items in Addition to the Items for Low Seismicity)** | | | | | | | |
| ***Seismic-Force-Resisting System*** | | | | | | | |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | REDUNDANCY: The number of lines of shear walls in each principal direction is greater than or equal to 2. | 5.5.1.1 | A.3.2.1.1 | |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | WALL SHEAR STRESS CHECK: The shear stress in the precast panels, calculated using the Quick Check procedure of Section 4.4.3.3 is less than the greater of 100 lb/in.$^2$ (0.69 MPa) or $2\sqrt{f'_c}$. | 5.5.3.1.1 | A.3.2.3.1 | |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | REINFORCING STEEL: The ratio of reinforcing steel area to gross concrete area is not less than 0.0012 in the vertical direction and 0.0020 in the horizontal direction. | 5.5.3.1.3 | A.3.2.3.2 | |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | WALL THICKNESS: Thicknesses of bearing walls are not less than 1/40 the unsupported height or length, whichever is shorter, nor less than 4 in. (101.6 mm). | 5.5.3.1.2 | A.3.2.3.5 | |
| ***Diaphragms*** | | | | | | | |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | TOPPING SLAB: Precast concrete diaphragm elements are interconnected by a continuous reinforced concrete topping slab with a minimum thickness of 2 in. (50.8 mm). | 5.6.4 | A.4.5.1 | |

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| | | | | | | |
|---|---|---|---|---|---|---|
| ***Connections*** | | | | | | |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | WOOD LEDGERS: The connection between the wall panels and the diaphragm does not induce cross-grain bending or tension in the wood ledgers or top plates fastened to the walls. | 5.7.1.3 | A.5.1.2 |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | TRANSFER TO SHEAR WALLS: Diaphragms are connected for transfer of seismic forces to the shear walls. | 5.7.2 | A.5.2.1 |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | TOPPING SLAB TO WALLS OR FRAMES: Reinforced concrete topping slabs that interconnect the precast concrete diaphragm elements are doweled for transfer of forces into the shear wall or frame elements. | 5.7.2 | A.5.2.3 |
| **C** ☐ | **NC** ☐ | **N/A** ☐ | **U** ☐ | GIRDER–COLUMN CONNECTION: There is a positive connection using plates, connection hardware, or straps between the girder and the column support. | 5.7.4.1 | A.5.4.1 |

| **Status** | **Evaluation Statement** | **Tier 2 Reference** | **Commentary Reference** | **Comments** |
|---|---|---|---|---|
| **High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)** | | | | |
| ***Seismic-Force-Resisting System*** | | | | |
| **C** ☐ **NC** ☐ **N/A** ☐ **U** ☐ | DEFLECTION COMPATIBILITY FOR RIGID DIAPHRAGMS: Secondary components have the shear capacity to develop the flexural strength of the components. | 5.5.2.5.2 | A.3.1.6.2 | |
| **C** ☐ **NC** ☐ **N/A** ☐ **U** ☐ | WALL OPENINGS: The total combined width of openings and wall piers with aspect ratios greater than 2-to-1 along any perimeter wall line constitutes less than 75% of the total length of any perimeter wall. | 5.5.3.3.1 | A.3.2.3.3 | |
| ***Diaphragms*** | | | | |
| **C** ☐ **NC** ☐ **N/A** ☐ **U** ☐ | DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations. | 5.6.1.1 | A.4.1.1 | |
| **C** ☐ **NC** ☐ **N/A** ☐ **U** ☐ | ROOF CHORD CONTINUITY: All chord elements are continuous, regardless of changes in roof elevation. | 5.6.1.1 | A.4.1.3 | |

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| C | NC | N/A | U | | | |
|---|---|---|---|---|---|---|
| ☐ | ☐ | ☐ | ☐ | CROSSTIES IN FLEXIBLE DIAPHRAGMS: There are continuous crossties between diaphragm chords to distribute the out-of-plane wall anchorage forces into the diaphragm. Where each out-of-plane connection does not have a continuous crosstie across the entire diaphragm, these connections are developed into subdiaphragms between crossties with a maximum length-to-width ratio of 3-to-1. | 5.6.1.2 | A.4.1.2 |
| C | NC | N/A | U | | | |
| ☐ | ☐ | ☐ | ☐ | STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 24 ft (7.3 m) and aspect ratios less than 2-to-1 in the direction being considered. | 5.6.2 | A.4.2.1 |
| C | NC | N/A | U | | | |
| ☐ | ☐ | ☐ | ☐ | DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 40 ft (12.2 m) and aspect ratios less than or equal to 4-to-1. | 5.6.2 | A.4.2.2 |
| C | NC | N/A | U | | | |
| ☐ | ☐ | ☐ | ☐ | BLOCKED DIAPHRAGMS: All blocked wood structural panel diaphragms have horizontal spans less than 120 ft (36.5 m) and have aspect ratios less than or equal to 4-to-1. | 5.6.2 | A.4.2.3 |
| C | NC | N/A | U | | | |
| ☐ | ☐ | ☐ | ☐ | CANTILEVERED WOOD DIAPHRAGMS: All cantilevered wood diaphragms that provide lateral support for concrete or masonry walls consist of wood structural panels and have a maximum cantilever length of 20 ft (6.1 m) if unblocked or 35 ft (10.7 m) if blocked, and a maximum ratio of cantilever length to diaphragm width of 1:2 if unblocked and 1:1 if blocked. In addition, the cantilevered diaphragm has a back-span length equal to or greater than the cantilevered portion. | 5.6.2 | A.4.2.4 |
| C | NC | N/A | U | | | |
| ☐ | ☐ | ☐ | ☐ | NON-CONCRETE-FILLED DIAPHRAGMS: Bare steel deck diaphragms or steel deck diaphragms with fill other than reinforced structural concrete consist of horizontal spans of less than 120 ft (36.5 m) and have aspect ratios less than 4-to-1. | 5.6.3 | A.4.3.1 |

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| C | NC | N/A | U | | | |
|---|---|---|---|---|---|---|
| ☐ | ☐ | ☐ | ☐ | OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing. | 5.6.5 | A.4.7.1 |
| ***Connections*** | | | | | | |
| **C** | **NC** | **N/A** | **U** | | | |
| ☐ | ☐ | ☐ | ☐ | MINIMUM NUMBER OF WALL ANCHORS PER PANEL: There are at least two anchors connecting each precast wall panel to the diaphragm elements. | 5.7.1.4 | A.5.1.3 |
| **C** | **NC** | **N/A** | **U** | | | |
| ☐ | ☐ | ☐ | ☐ | PRECAST WALL PANELS: Precast wall panels are connected to the foundation. | 5.7.3.4 | A.5.3.6 |
| **C** | **NC** | **N/A** | **U** | | | |
| ☐ | ☐ | ☐ | ☐ | UPLIFT AT PILE CAPS: Pile caps have top reinforcement, and piles are anchored to the pile caps. | 5.7.3.5 | A.5.3.8 |
| **C** | **NC** | **N/A** | **U** | | | |
| ☐ | ☐ | ☐ | ☐ | GIRDERS: Girders supported by walls or pilasters have at least two ties securing the anchor bolts unless provided with independent stiff wall anchors with strength to resist the connection force calculated in the Quick Check procedure of Section 4.4.3.7. | 5.7.4.2 | A.5.4.2 |

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